



RNA-seq DESeq2 Differential Expression and QC Workflow

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Overview

This workflow demonstrates a reproducible RNA-seq differential expression and reporting pipeline using the public GEO dataset GSE203539.

Raw sequencing data were originally processed through a Galaxy-based RNA-seq pipeline including quality assessment, read trimming, alignment to the GRCm39 reference genome, and feature-level quantification prior to downstream analysis.

This repository focuses on downstream quality control, outlier sensitivity assessment, DESeq2 differential expression analysis, and structured result export workflows in R.

Analysis Plan

1. Load RNA-seq count data from `data/all_counts.txt`
2. Parse sample metadata
3. Perform QC and exploratory diagnostics on all samples
4. Evaluate pre-identified outlier samples through sensitivity analysis
5. Run filtered DESeq2 differential expression analysis
6. Export QC figures, DEG tables, and summary outputs to `results/`

Load packages

Project paths

Import count matrix

Build sample metadata

```
outlier_samples <- c("AEC.mock.III", "AEC.WSN.PP1.I", "AM.mock.II", "AM.WSN.III")

metadata <- tibble(sample_id = colnames(count_matrix)) %>%
  mutate(
    cell_type = case_when(
      str_starts(sample_id, "AEC\\.") ~ "AEC",
      str_starts(sample_id, "AM\\.") ~ "AM",
      TRUE ~ NA_character_
    ),
    condition = case_when(
      str_detect(sample_id, "\\..mock\\.") ~ "mock",
      str_detect(sample_id, "\\..WSN\\.PP1\\.") ~ "WSN_PP1",
      str_detect(sample_id, "\\..WSN\\.PP\\.") ~ "WSN_PP1",
      str_detect(sample_id, "\\..WSN\\.") ~ "WSN",
      TRUE ~ NA_character_
    ),
    replicate = str_extract(sample_id, "[^.]+"),
    outlier_flag = sample_id %in% outlier_samples
  ) %>%
  mutate(
    cell_type = factor(cell_type, levels = c("AEC", "AM")),
    condition = factor(condition, levels = c("mock", "WSN", "WSN_PP1")),
    replicate = factor(replicate),
    group = factor(paste(cell_type, condition, sep = "."))
  )

if (any(is.na(metadata$cell_type)) || any(is.na(metadata$condition))) {
  print(metadata)
  stop("Unable to infer cell_type or condition for one or more samples.
       Check sample naming conventions.")
}

missing_outliers <- setdiff(outlier_samples, metadata$sample_id)

if (length(missing_outliers) > 0) {
  warning("Pre-specified outlier sample(s) not found in count matrix: ",
         paste(missing_outliers, collapse = ", "))
}

if (!all(metadata$sample_id == colnames(count_matrix))) {
  stop("Metadata sample order does not match count matrix column order.")
}

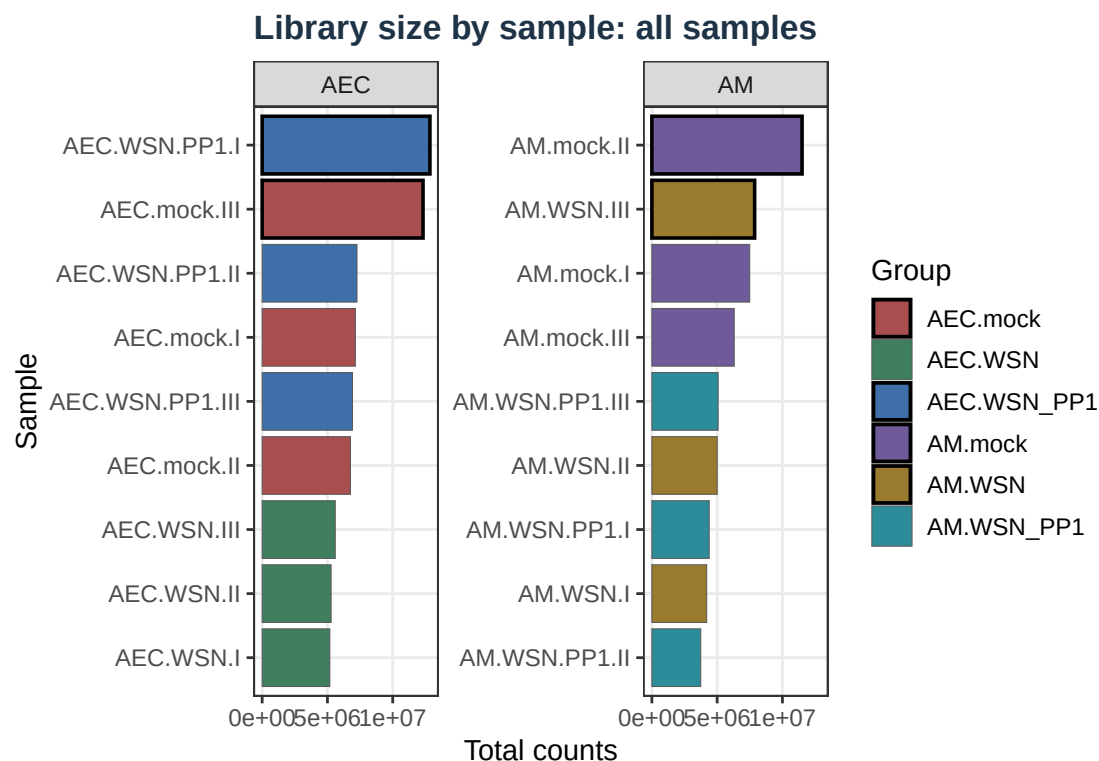
write_csv(metadata, file.path(qc_dir, "sample_metadata_with_outlier_flags.csv"))
```

Optional gene annotation

QC on all samples

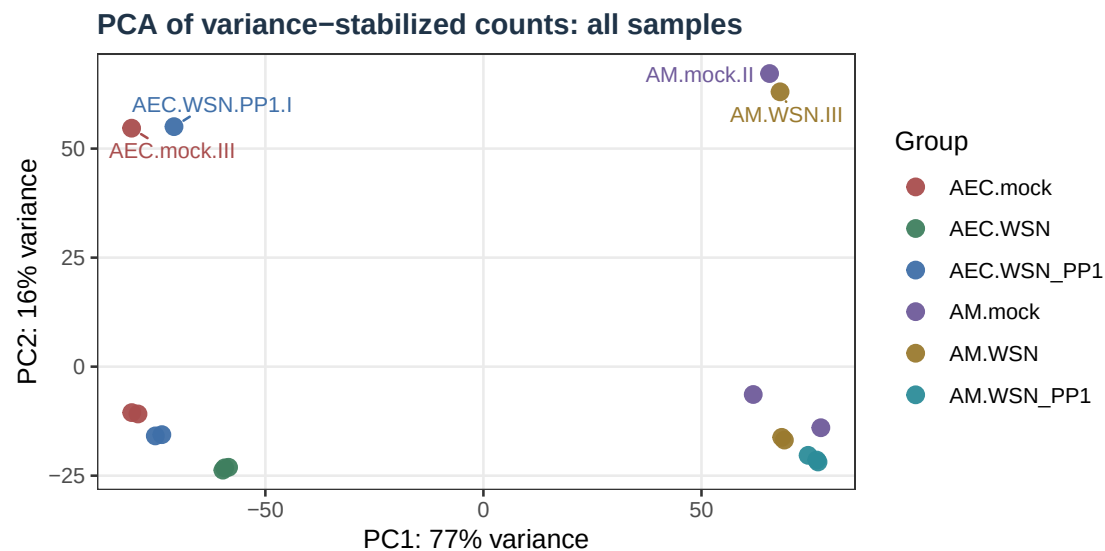
Quality control is performed before excluding any samples. The outlier samples are flagged visually and retained in this section.

Library size

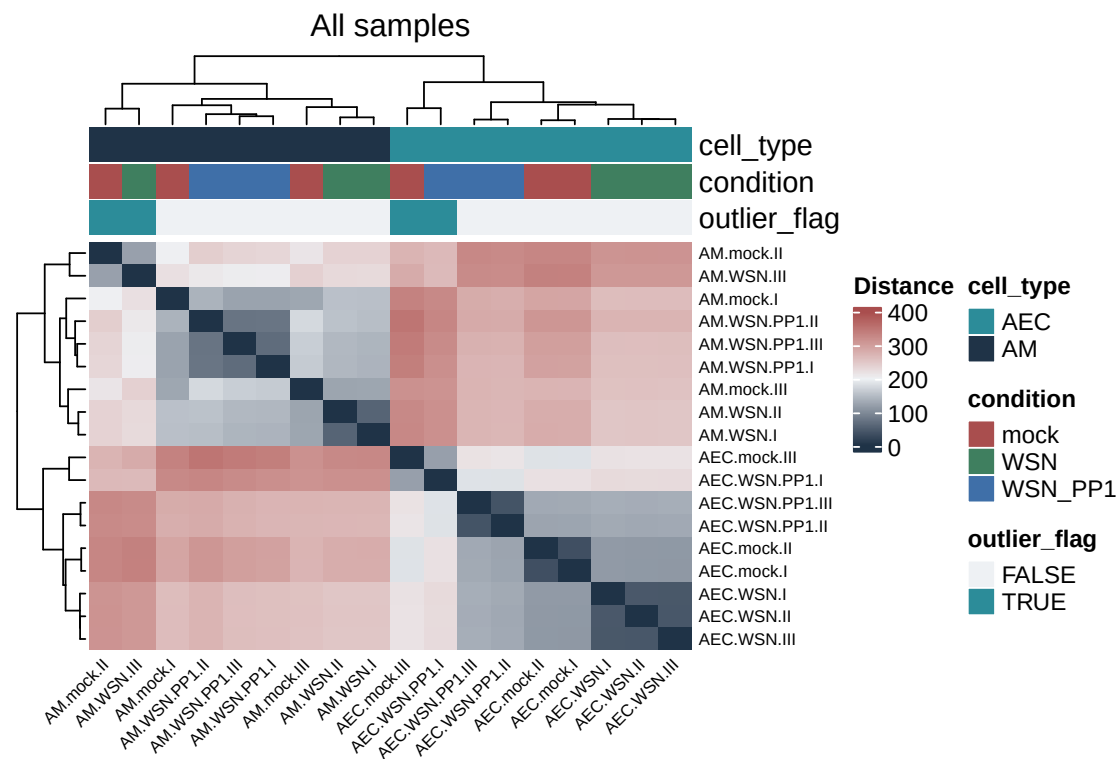


Variance-stabilized counts

PCA: all samples



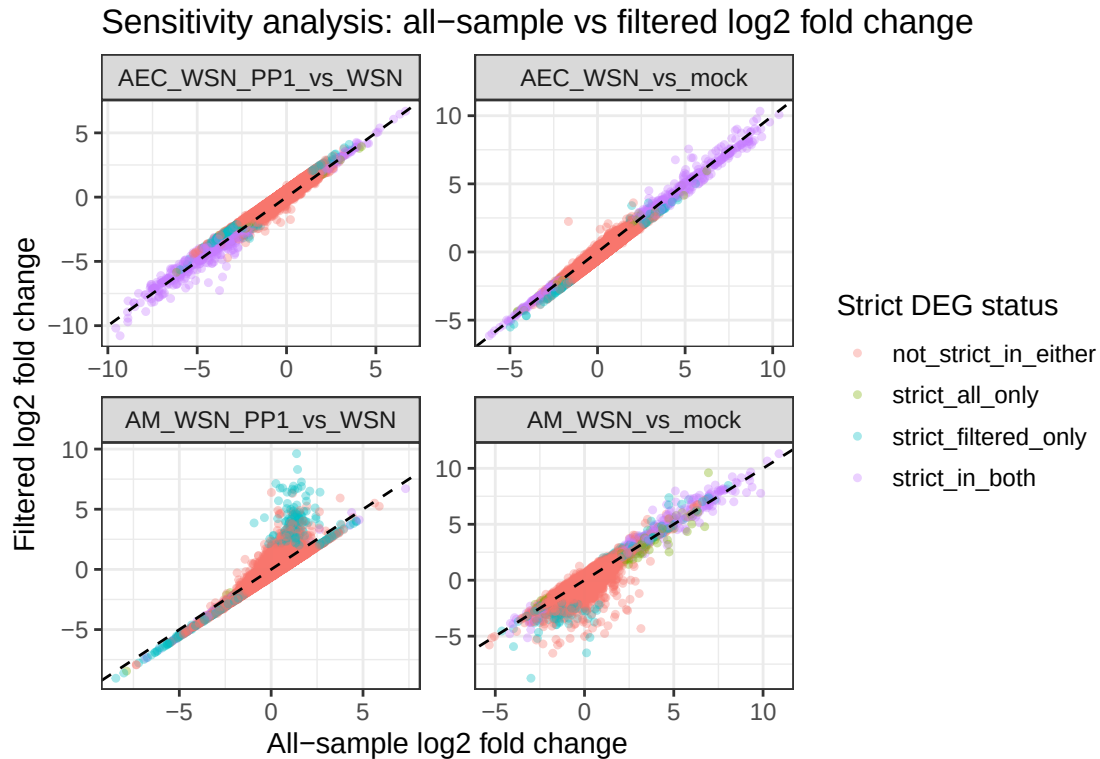
Sample distance heatmap: all samples



Helper functions

Sensitivity analysis: all-sample vs filtered DESeq2 results

This section evaluates whether the pre-specified outlier handling decision changes differential expression outputs. Both all-sample and filtered results are exported for transparency.

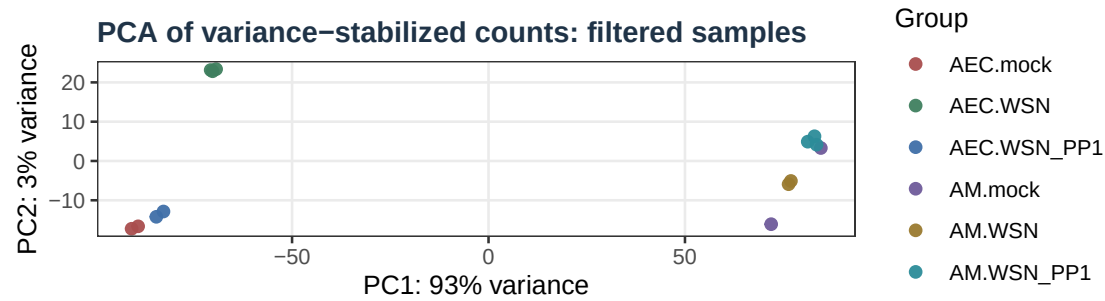


The sensitivity analysis shows that most high-confidence DEG calls are stable across filtering decisions, while selected contrasts, especially AM inhibitor-response comparisons, show larger shifts after outlier removal. This supports retaining both all-sample and filtered outputs for transparency.

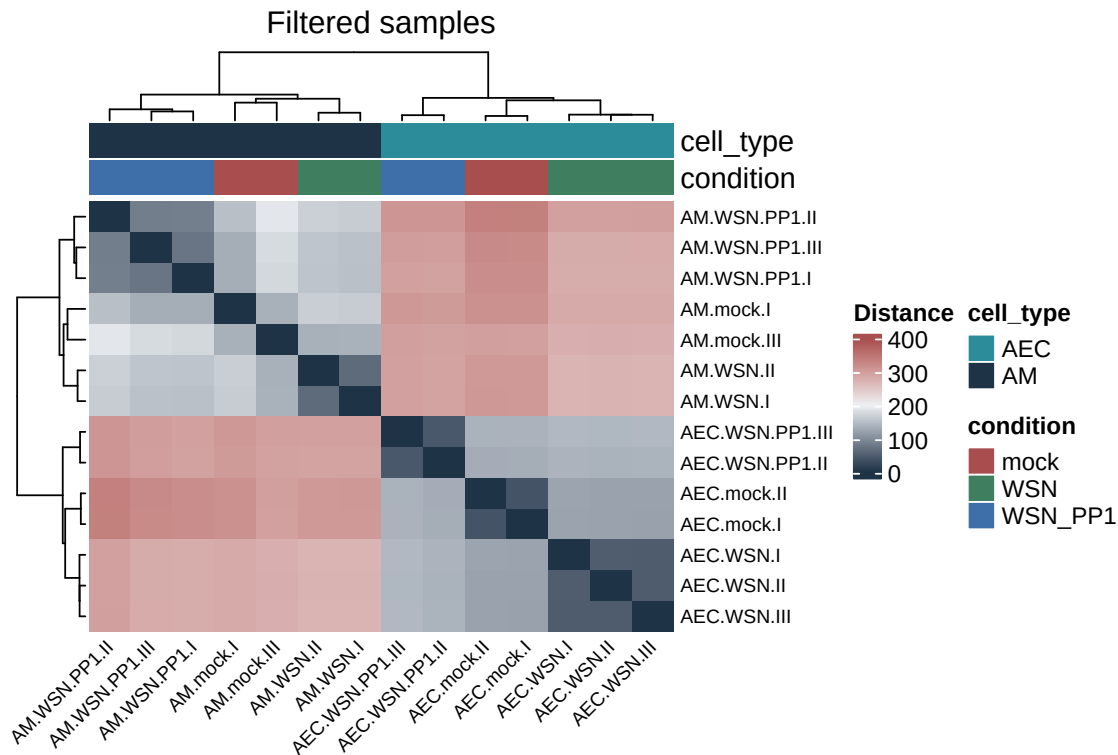
Filtered DESeq2 workflow

The remaining workflow uses the filtered dataset, excluding the explicitly flagged outlier samples.

Filtered PCA



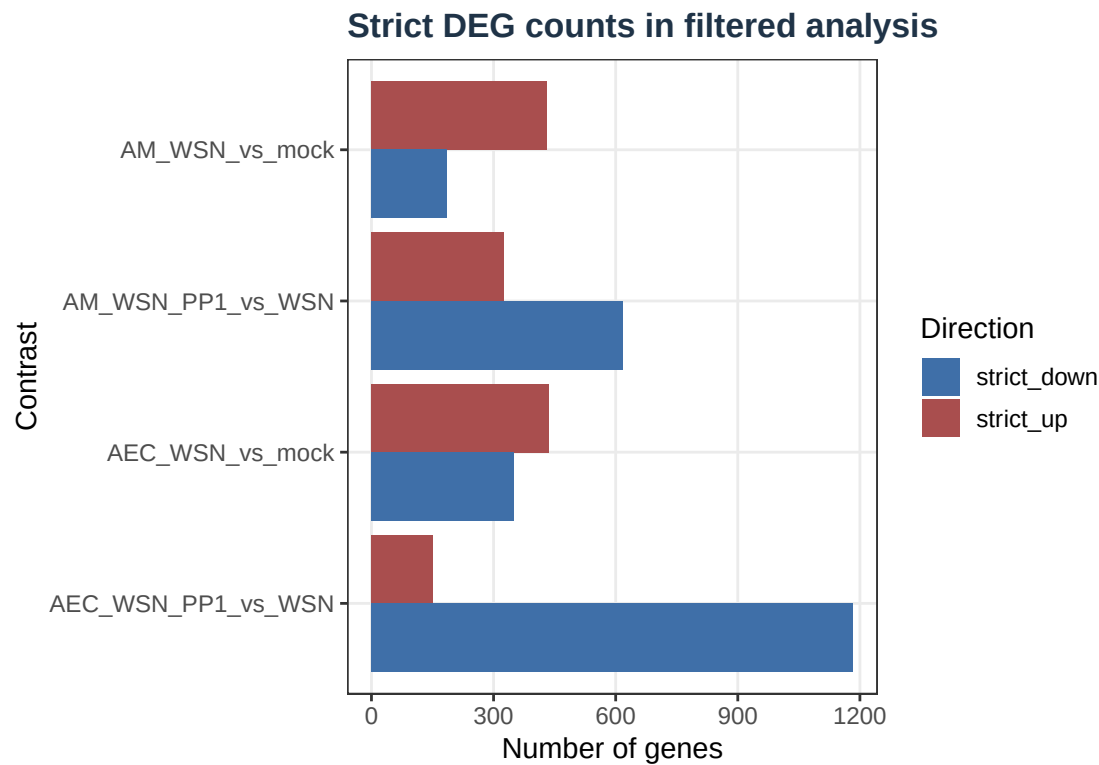
Filtered sample distance heatmap



Export filtered DEG results

Filtered DEG visualizations

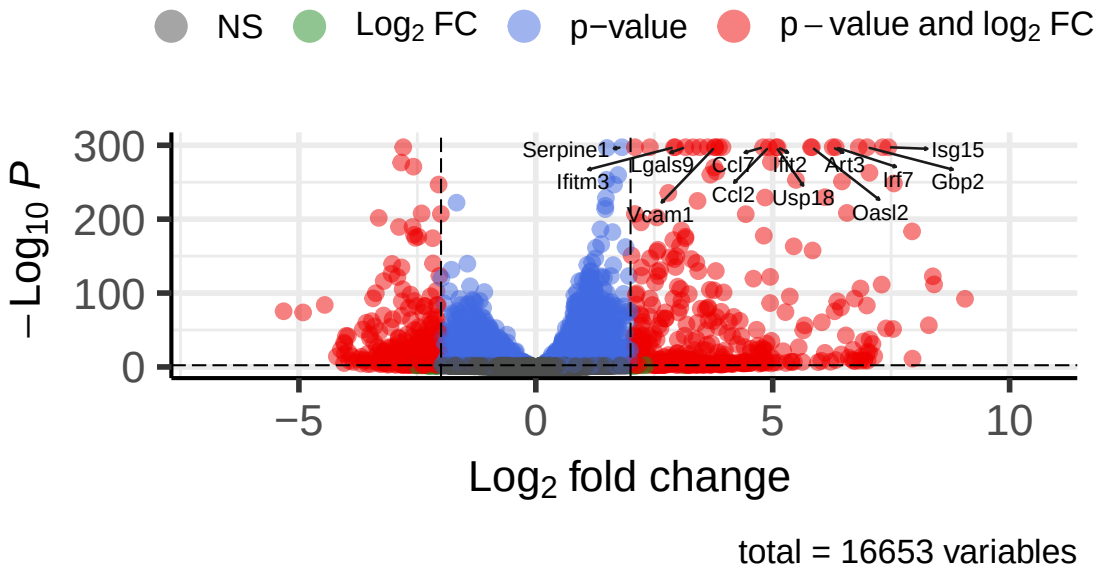
DEG count summary



EnhancedVolcano plots with gene labels

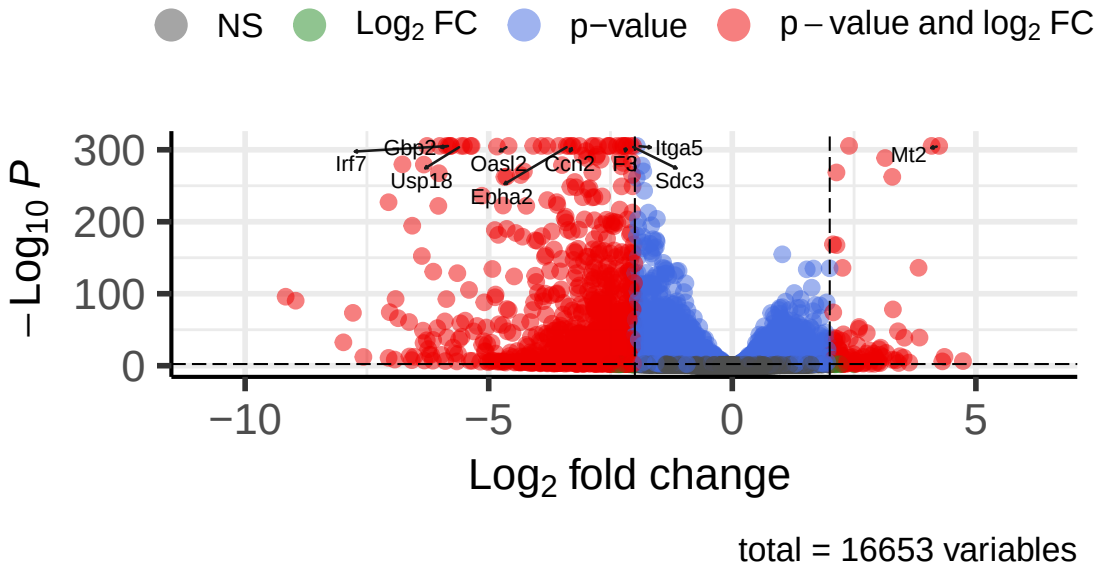
AEC_WSN_vs_mock

WSN vs mock



AEC_WSN_PP1_vs_WSN

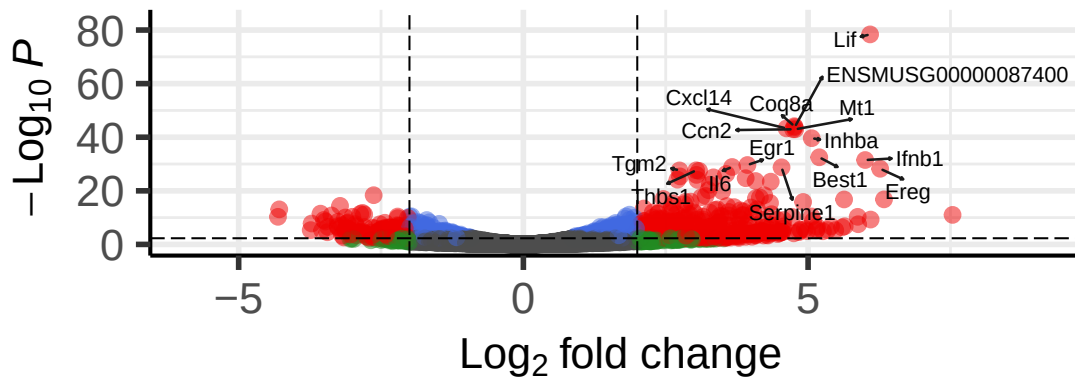
WSN_PP1 vs WSN



AM_WSN_vs_mock

WSN vs mock

● NS ● Log₂ FC ● p-value ● p-value and log₂ FC

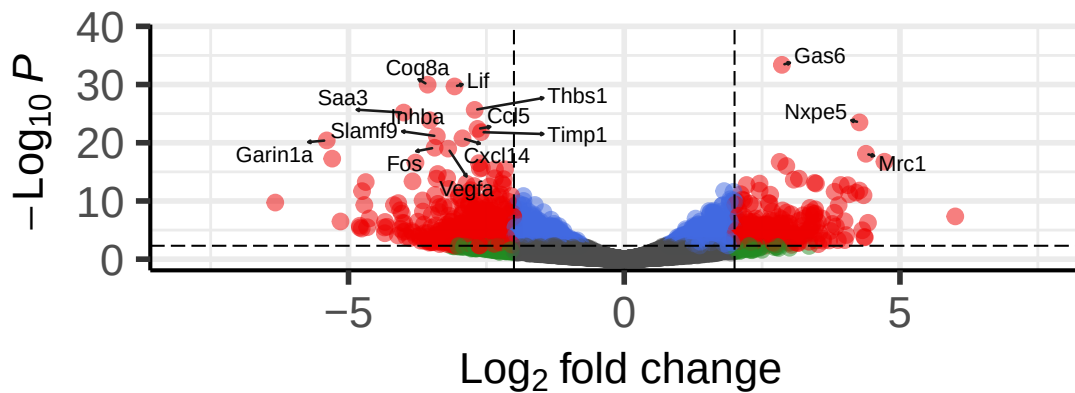


total = 14777 variables

AM_WSN_PP1_vs_WSN

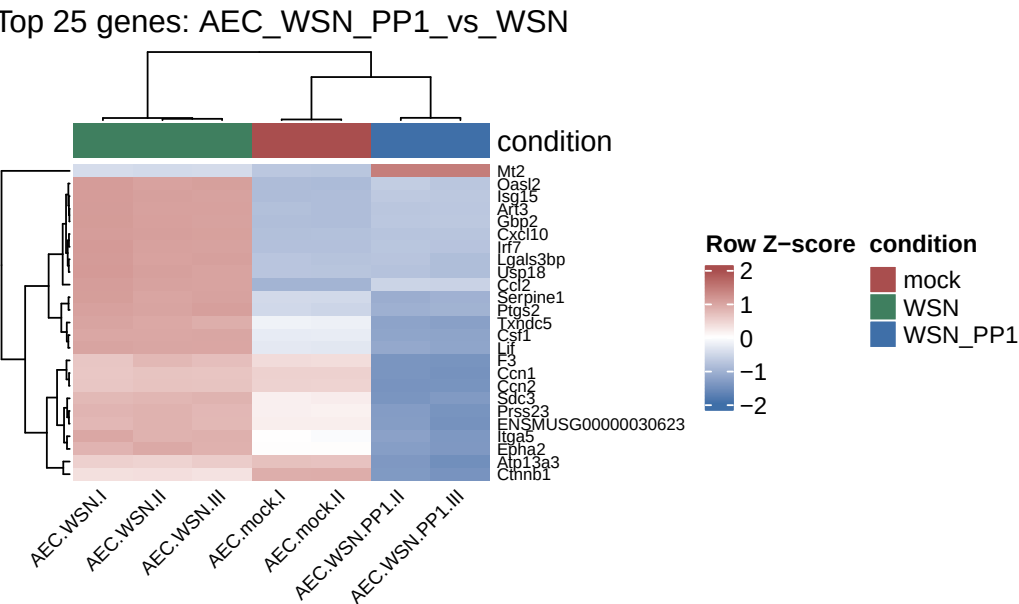
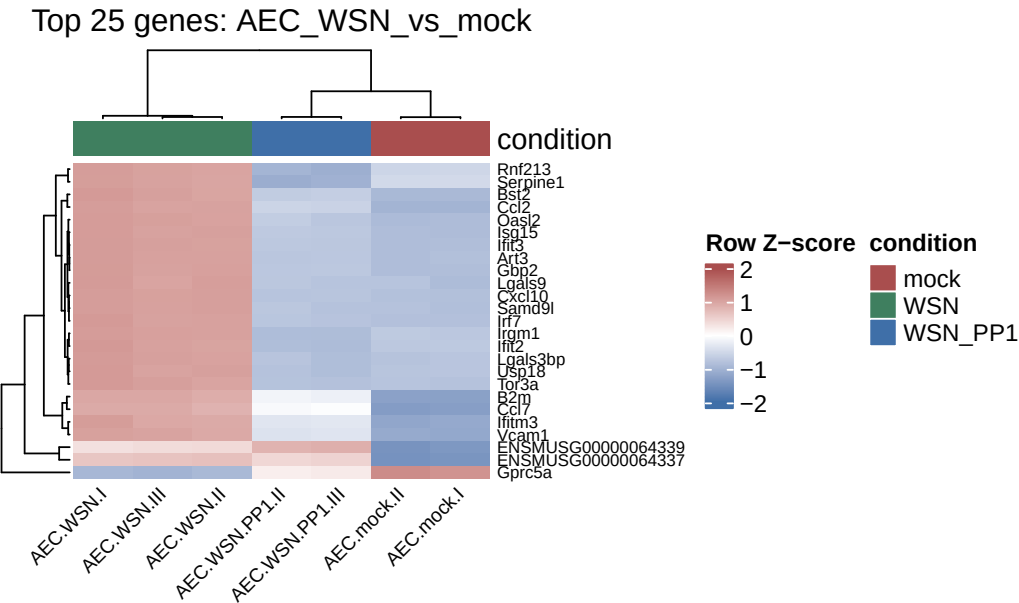
WSN_PP1 vs WSN

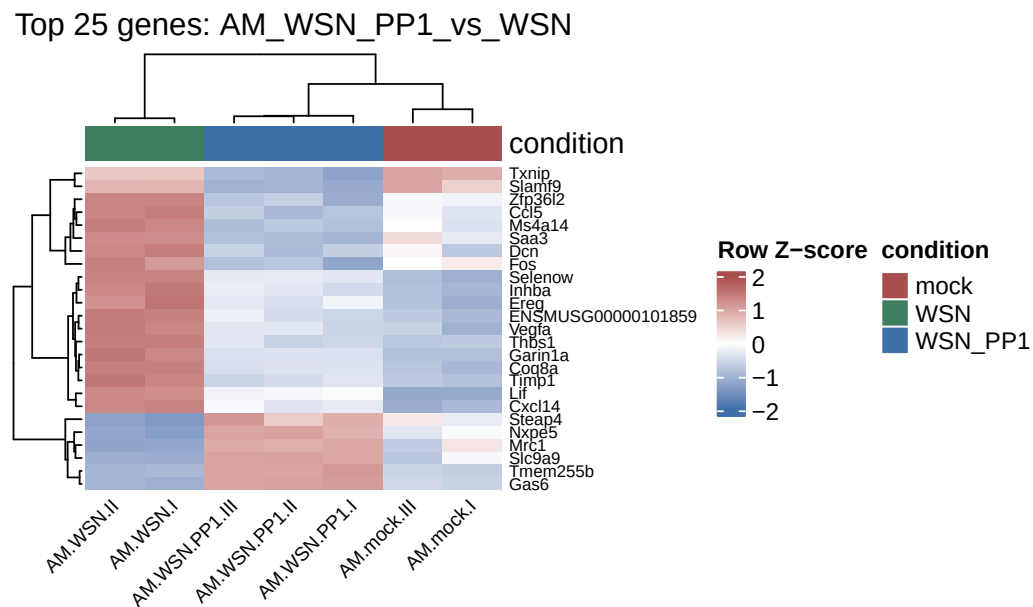
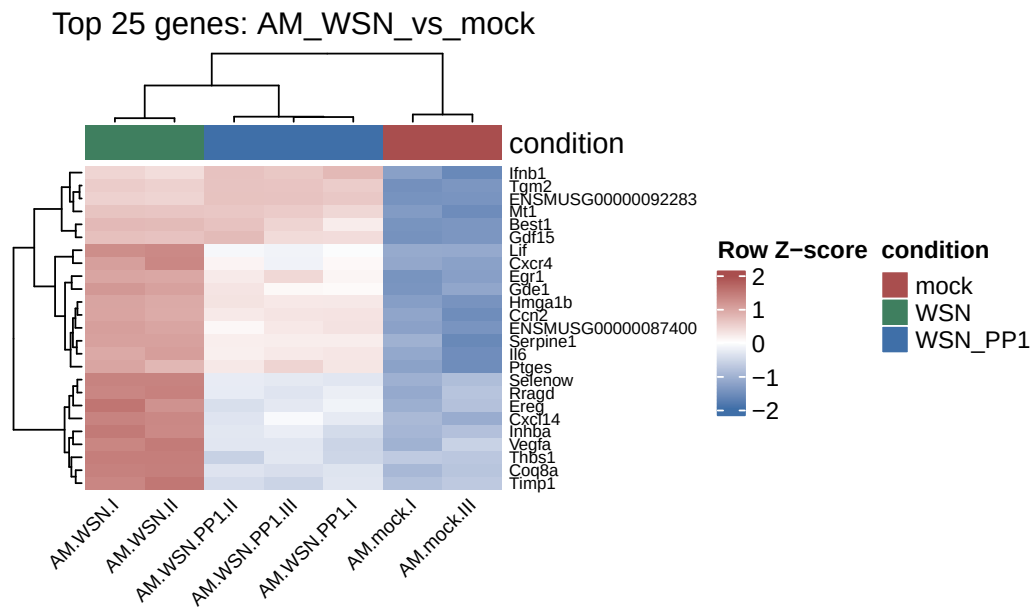
● NS ● Log₂ FC ● p-value ● p-value and log₂ FC



total = 14777 variables

Heatmaps of top DEGs





Conclusions

This workflow demonstrates a reproducible RNA-seq analysis and reporting pipeline integrating:

- QC and exploratory diagnostics
- transparent outlier sensitivity analysis
- DESeq2 differential expression workflows
- structured export of annotated DEG outputs
- downstream-ready reporting artifacts for Python-based aggregation and visualization

The analysis emphasizes reproducibility and transparent analytical decision-making rather than biological overinterpretation.

Reproducibility information

```
sessionInfo()
```

```
## R version 4.4.0 (2024-04-24 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26200)
##
## Matrix products: default
##
## locale:
## [1] LC_COLLATE=English_United States.utf8
## [2] LC_CTYPE=English_United States.utf8
## [3] LC_MONETARY=English_United States.utf8
## [4] LC_NUMERIC=C
## [5] LC_TIME=English_United States.utf8
##
## time zone: America/Los_Angeles
## tzcode source: internal
##
## attached base packages:
## [1] grid      stats4    stats      graphics  grDevices  utils      datasets
## [8] methods  base
##
## other attached packages:
## [1] purrr_1.2.1          tibble_3.3.1
## [3] stringr_1.6.0        readr_2.2.0
## [5] tidyr_1.3.2          dplyr_1.2.0
## [7] pheatmap_1.0.13      org.Mm.eg.db_3.20.0
## [9] AnnotationDbi_1.68.0 circlize_0.4.18
## [11] ComplexHeatmap_2.22.0 EnhancedVolcano_1.24.0
## [13] ggrepel_0.9.8         ggplot2_4.0.2
## [15] DESeq2_1.46.0         SummarizedExperiment_1.36.0
## [17] Biobase_2.66.0        MatrixGenerics_1.18.1
## [19] matrixStats_1.5.0     GenomicRanges_1.58.0
## [21] GenomeInfoDb_1.42.3   IRanges_2.40.1
## [23] S4Vectors_0.44.0      BiocGenerics_0.52.0
##
## loaded via a namespace (and not attached):
## [1] DBI_1.3.0             rlang_1.1.7           magrittr_2.0.4
## [4] clue_0.3-68           GetoptLong_1.1.1       otel_0.2.0
## [7] compiler_4.4.0        RSQLite_2.4.6          png_0.1-9
## [10] systemfonts_1.3.2     vctrs_0.7.2           pkgconfig_2.0.3
```

## [13] shape_1.4.6.1	crayon_1.5.3	fastmap_1.2.0
## [16] XVector_0.46.0	labeling_0.4.3	utf8_1.2.6
## [19] rmarkdown_2.31	tzdb_0.5.0	UCSC.utils_1.2.0
## [22] ragg_1.5.2	bit_4.6.0	xfun_0.57
## [25] zlibbioc_1.52.0	cachem_1.1.0	jsonlite_2.0.0
## [28] blob_1.3.0	DelayedArray_0.32.0	BiocParallel_1.40.0
## [31] parallel_4.4.0	cluster_2.1.6	R6_2.6.1
## [34] stringi_1.8.7	RColorBrewer_1.1-3	Rcpp_1.1.1
## [37] iterators_1.0.14	knitr_1.51	Matrix_1.7-0
## [40] tidyselect_1.2.1	rstudioapi_0.18.0	abind_1.4-8
## [43] yaml_2.3.12	doParallel_1.0.17	codetools_0.2-20
## [46] lattice_0.22-6	withr_3.0.2	KEGGREST_1.46.0
## [49] S7_0.2.1	evaluate_1.0.5	Biostrings_2.74.1
## [52] pillar_1.11.1	foreach_1.5.2	generics_0.1.4
## [55] vroom_1.7.0	hms_1.1.4	scales_1.4.0
## [58] glue_1.8.0	tools_4.4.0	locfit_1.5-9.12
## [61] colorspace_2.1-2	GenomeInfoDbData_1.2.13	cli_3.6.2
## [64] textshaping_1.0.5	S4Arrays_1.6.0	gtable_0.3.6
## [67] digest_0.6.39	SparseArray_1.6.2	rjson_0.2.23
## [70] farver_2.1.2	memoise_2.0.1	htmltools_0.5.9
## [73] lifecycle_1.0.5	httr_1.4.8	GlobalOptions_0.1.4
## [76] bit64_4.6.0-1		